# The Normal Distribution

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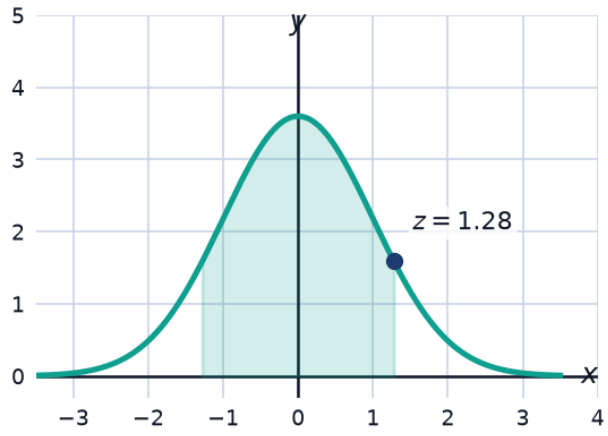
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Properties of the normal distribution

State three key properties of a normal distribution's shape.

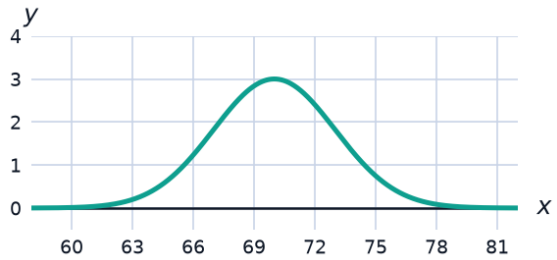
**2** [Foundational] • Target: Properties of the normal distribution

Explain what the two parameters $\mu$ and $\sigma$ control in a normal distribution's shape.




**3** [Foundational] • Target: The 68-95-99.7 rule

Adult male heights are approximately normal with mean $70$ inches and standard deviation $3$ inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.




**4** [Foundational] • Target: The 68-95-99.7 rule

Using the same distribution, find the approximate percentage of adult men taller than $76$ inches.

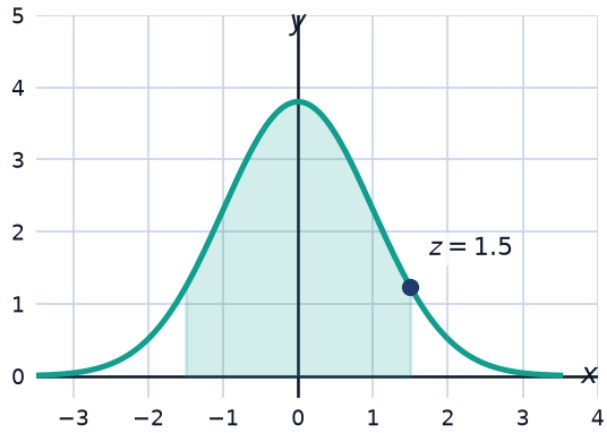
**5** [Foundational] • Target: Standardizing and using the normal distribution

Using the same height distribution (mean $70$, standard deviation $3$), find the $z$-score for a height of $77.5$ inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than $77.5$ inches.

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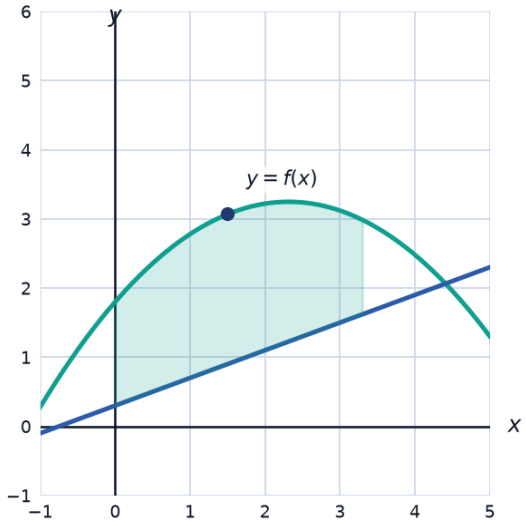
**6** [Foundational] • Target: Standardizing and using the normal distribution

Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.




**7** [Foundational] • Target: Visualizing the standard normal distribution

The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).




**a** Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.

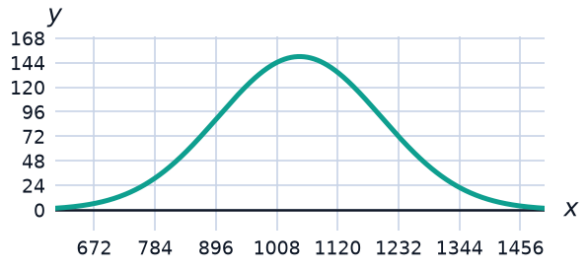
**b** Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Working backward: finding values from percentiles

SAT scores are approximately normal with mean $1050$ and standard deviation $150$. A student scores at the $90$th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.

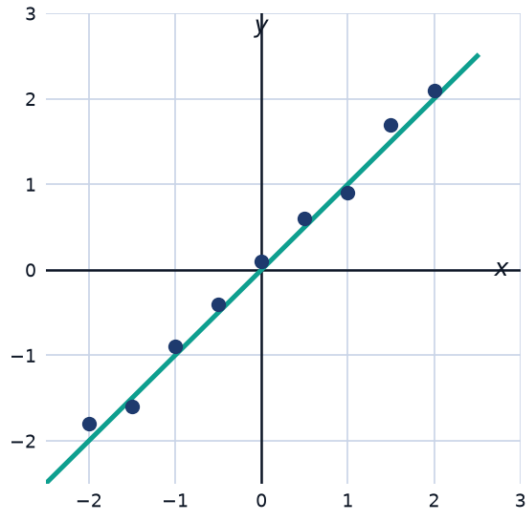



**9** [Intermediate] • Target: Working backward: finding values from percentiles

Using the same distribution, find the score that separates the bottom $10\%$ of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.

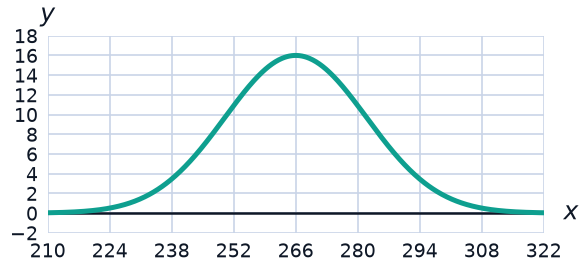
**10** [Intermediate] • Target: Assessing normality

Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.




**11** [Intermediate] • Target: Applying the normal distribution to real data

Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the $z$-score for a pregnancy lasting $250$ days.




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**12** [Intermediate] • Target: Applying the normal distribution to real data

Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between $234$ and $298$ days.

**13** [Intermediate] • Target: Combining two independent normal random variables

A vending machine dispenses a soda whose volume is normally distributed with mean $355$ mL and standard deviation $4$ mL, independently of a snack bag whose weight is normally distributed with mean $50$ g and standard deviation $2$ g. If $T$ represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of $T$. (This is a simplified illustrative combination, not a real physical unit.)

**14** [Intermediate] • Target: Combining two independent normal random variables

Battery lifespans are normally distributed with mean $500$ hours and standard deviation $40$ hours. Find the $z$-score for a battery lasting $460$ hours, and describe its relative position.

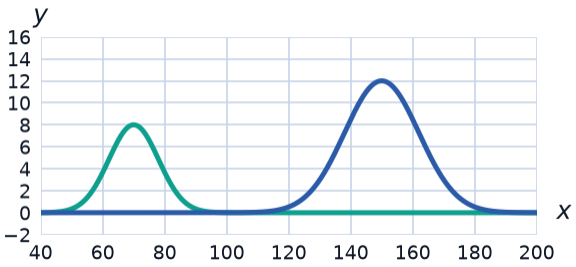
## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A company manufactures resistors with resistance values approximately normal, mean $100$ ohms and standard deviation $5$ ohms. A resistor is considered defective if its resistance is below $90$ ohms or above $110$ ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

**16** [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using $z$-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.




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**17** [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 2.01 Sampling Methods & Bias

The distribution of cholesterol levels in a population is approximately normal with $\mu = 190$ mg/dL and $\sigma = 24$ mg/dL. (a) What proportion of individuals have cholesterol levels above 226 mg/dL? (b) What cholesterol level marks the 90th percentile?

**18** [Challenging] • Target: The 68-95-99.7 rule • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A basketball player makes 75% of free throws. In a game, she shoots 12 free throws. Assuming attempts are independent: (a) Find the probability she makes exactly 9 free throws. (b) Find the mean and standard deviation of made free throws.

**19** [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**20** [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Properties of the normal distribution

A normal distribution has mean $64$ and standard deviation $9$. Using the 68-95-99.7 rule, find the range containing the middle $95\%$ of values.

**a.** $(46, 82)$

**b.** $(37, 91)$

**c.** $(46, 81)$

**d.** $(47, 82)$

**22** [Foundational] • Target: The 68-95-99.7 rule

A normal distribution has mean $\mu = 101$ and standard deviation $\sigma = 4$. Find the approximate percentage of values below $x = 103$, to two decimal places.

**a.** 69.15%

**b.** 72.15%

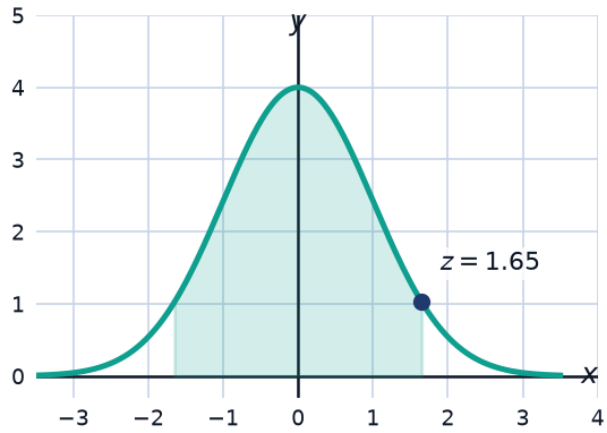
**c.** 66.15%

**d.** 30.85%

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**23** [Foundational] • Target: Standardizing and using the normal distribution

A normal distribution has mean $\mu = 144$ and standard deviation $\sigma = 16$. Find the approximate percentage of values between $x_1 = 136$ and $x_2 = 152$, to two decimal places.




**a.** 69.15%

**b.** 34.29%

**c.** 42.29%

**d.** 38.29%

**24** [Foundational] • Target: Visualizing the standard normal distribution

Scores on an exam are approximately normal with mean $\mu = 390$ and standard deviation $\sigma = 15$. A student scores at the $95$th percentile, corresponding to $z \approx 1.645$. Find that student's approximate score.

**a.** 365

**b.** 405

**c.** 425

**d.** 415

**25** [Foundational] • Target: Working backward: finding values from percentiles

A company manufactures parts with a measurement approximately normal, mean $152$ and standard deviation $5$. A part is considered defective if its measurement is below $142$ or above $162$ (i.e. outside $\mu \pm 2\sigma$). Using the 68-95-99.7 rule, estimate the percentage of parts that are defective.

**a.** 0.3%

**b.** 5%

**c.** 32%

**d.** 2.5%

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**26** [Foundational] • Target: Assessing normality

A data set has $Q_1 = 17$ and $Q_3 = 35$. Using the 1.5×IQR rule, which of the following values would be classified as an outlier?

**a.** 35

**b.** -14

**c.** 17

**d.** -9

**27** [Foundational] • Target: Applying the normal distribution to real data

A researcher is designing an experiment to test the effectiveness of a new fertilizer on plant growth. They have a total of 50 experimental plots available. The experiment will involve two treatments: the new fertilizer (Treatment A) and a standard fertilizer (Treatment B). To ensure fair comparison and control for confounding variables, they decide to use a randomized block design. The plots are divided into 5 blocks based on soil quality, with each block containing 10 plots. Within each block, the plots will be randomly assigned to either Treatment A or Treatment B. If the number of plots assigned to Treatment A within the first block is $x$, and the total number of plots assigned to Treatment A across all 5 blocks is 25, and the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1, what is the value of $x$?

**a.** 3

**b.** 4

**c.** 5

**d.** 6

**28** [Foundational] • Target: Combining two independent normal random variables

Two independent random samples, each of size $n$, are taken from a normal population with mean $\mu$ and known standard deviation $\sigma$. From each sample, a symmetric $90\%$ confidence interval for $\mu$ is constructed. Given that $\Phi(1.645) = 0.9500$ and $\Phi(2.326) = 0.9900$, where $\Phi$ denotes the standard normal cumulative distribution function, find the probability that these two confidence intervals overlap.

**a.** 0.980

**b.** 0.900

**c.** 0.810

**d.** 0.990

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Properties of the normal distribution

A researcher is designing a survey to understand student opinions on campus dining. The researcher decides to select students from the student directory using a systematic random sampling method. If the student directory has a total of $1500$ students, and the researcher decides to select every $10$th student, starting with a randomly selected student between 1 and 10, how many students will be in the sample?

**a.** 150

**b.** 10

**c.** 15

**d.** 1500

**30** [Intermediate] • Target: The 68-95-99.7 rule

A data set contains the values 4, 13, 14, 23, 30, 33, 35. What is the mean of this data set?

**a.** 23

**b.** 21.71

**c.** 23.89

**d.** 19.71

**31** [Intermediate] • Target: Standardizing and using the normal distribution

A data set contains the values 22, 25, 41, 51, 54, 64. What is the mean of this data set (rounded to 2 decimal places)?

**a.** 48.83

**b.** 38.83

**c.** 42.83

**d.** 45.83

**32** [Intermediate] • Target: Visualizing the standard normal distribution

A researcher wants to investigate the relationship between the type of pet owned (Dog, Cat, Fish) and the preferred music genre (Rock, Pop, Classical). They collect data and present it in a two-way table. If the table has 3 rows and 3 columns, what is the total number of cells in the table that represent a combination of pet ownership and music genre?

**a.** 6

**b.** 7

**c.** 9

**d.** 16

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**33** [Intermediate] • Target: Working backward: finding values from percentiles

A researcher is investigating the relationship between study habits and exam performance. They categorize students into two groups: 'Regular Study' and 'Cramming'. They also categorize exam performance into 'Pass' and 'Fail'. The data collected is presented in a two-way table. If the table has 2 rows representing study habits and 3 columns representing exam performance categories, what is the total number of cells in this two-way table?

**a.** 4

**b.** 5

**c.** 6

**d.** 3

**34** [Intermediate] • Target: Assessing normality

The IQR (interquartile range) equals:

**a.** $Q_3 - Q_1$

**b.** $Q_1 - Q_3$

**c.** Max – Min

**d.** $Q_2 - Q_1$

**35** [Intermediate] • Target: Applying the normal distribution to real data

From a randomly ordered list of 1854 customers, a researcher selects every 21th person after choosing a random starting point. What sampling method is this?

**a.** Convenience sample

**b.** Systematic sample

**c.** Stratified random sample

**d.** Simple random sample (SRS)

**36** [Intermediate] • Target: Combining two independent normal random variables

From a randomly ordered list of 1002 employees, a researcher selects every 14th person after choosing a random starting point. What sampling method is this?

**a.** Cluster sample

**b.** Systematic sample

**c.** Voluntary response sample

**d.** Stratified random sample

**37** [Intermediate] • Target: Properties of the normal distribution

From a randomly ordered list of 885 voters, a researcher selects every 24th person after choosing a random starting point. What sampling method is this?

**a.** Simple random sample (SRS)

**b.** Voluntary response sample

**c.** Convenience sample

**d.** Systematic sample

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: The 68-95-99.7 rule • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

From a randomly ordered list of 610 students, a researcher selects every 18th person after choosing a random starting point. What sampling method is this?

**a.** Simple random sample (SRS)

**b.** Systematic sample

**c.** Convenience sample

**d.** Stratified random sample

**39** [Challenging] • Target: Standardizing and using the normal distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

From a randomly ordered list of 1390 athletes, a researcher selects every 7th person after choosing a random starting point. What sampling method is this?

**a.** Stratified random sample

**b.** Systematic sample

**c.** Simple random sample (SRS)

**d.** Voluntary response sample

**40** [Challenging] • Target: Visualizing the standard normal distribution • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

From a randomly ordered list of 1515 patients, a researcher selects every 6th person after choosing a random starting point. What sampling method is this?

**a.** Convenience sample

**b.** Systematic sample

**c.** Simple random sample (SRS)

**d.** Stratified random sample

**41** [Challenging] • Target: Working backward: finding values from percentiles • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A data set (already sorted, n=9) is 1, 7, 10, 11, 25, 34, 43, 46, 51. What is the five-number summary?

**a.** Min = 1, Q1 = 10, Median = 51, Q3 = 43, Max = 25

**b.** Min = 1, Q1 = 43, Median = 25, Q3 = 10, Max = 51

**c.** Min = 1, Q1 = 10, Median = 25, Q3 = 43, Max = 51

**d.** Min = 51, Q1 = 10, Median = 25, Q3 = 43, Max = 1

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**42** [Challenging] • Target: Assessing normality • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

An outlier in a boxplot is typically beyond:

**a.** 1.5 × IQR from Q1 or Q3

**b.** 2 × IQR

**c.** 3 standard deviations

**d.** The median

**43** [Challenging] • Target: Applying the normal distribution to real data • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A regression line predicts $\hat{y} = 41.85$ for $x = 25$, while the observed value is $y = 39.1$. What is the residual for this point?

**a.** 3.25

**b.** -2.75

**c.** -0.75

**d.** -6.75

**44** [Challenging] • Target: Combining two independent normal random variables • Cross-Lesson: 1.01 Measures of Center & Spread

Which test compares two independent population means?

**a.** One-sample z-test

**b.** Paired t-test

**c.** Two-sample t-test

**d.** Chi-square test

**45** [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A randomised block design blocks subjects by age group into 3 blocks of 44 subjects each, with random assignment to treatments within each block. What is the total sample size?

**a.** 117

**b.** 107

**c.** 142

**d.** 132

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